Chapter 13 Notes

Section 1

Notes

Note that, on page 170, the separation axioms are listed incorrectly. They should use the formula $\phi(y, y_1, ..., y_n)$, not $\phi(x, y_1, ..., y_n)$.

Exercise 1.1 (a)

We prove that K has property F_2' iff all the separation axioms are true over K.

(a) On the one hand, suppose that K has property F_2'. Now consider any set $x \in K$ and any formula $\phi(y, y_1, ..., y_n)$ (for some elements $y_1, ..., y_n$ of K). Then let A be the class of all y such that $\phi(y, y_1, ..., y_n)$ is true over K (A is defined by $\phi(y, y_1, ..., y_n)$ over K). By F_2', $A \cap x \in K$. But $A \cap x$ satisfies the formula in **Ax 2**. Thus, for each formula $\phi(y, y_1, ..., y_n)$ and $x \in K$, the separation axiom for that formula and element holds over K.

(b) On the other hand, suppose that all the separation axioms are true over K. Then consider any first-order definable subclass A of K and any element b of K. A must be definable over K by some formula $\phi(x)$. Then, by the separation axiom, there is some $z \in K$ such that, for all $x \in K$, $x \in z \equiv (x \in b \wedge \phi(x))$. This formula works as, if $x \in b \cap A$, then $x \in A$, and thus $x \in K$, meaning the formula would hold true. Of course, if the formula is true, then $x \in b \cap A$.

Exercise 1.1 (b)

We prove that K (where K is transitive) is a first-order universe iff all the axioms of ZF are true over K.

(a) On the one hand, suppose that K is a first-order universe. We show that all the ZF axioms are true over K.

Ax 1: Suppose that $(\forall z)(z \in x \equiv z \in y)$. Then, because K is transitive (by F_1), it follows that $x = y$.

Ax 2: We show that this holds by showing that F_2' holds (in K), and consequently **Ax 2** would hold by Exercise 1.1 (a). So suppose A is a first-order definable subclass of K and b is an element of K. Then, because K is first-order swelled, it follows that $A \cap b \in K$ (because $A \cap b$ is first-order definable over K, as it is defined by the formula $(y \in z \equiv (y \in x \wedge \phi(y)))$ where $\phi(y)$ is some formula that defines A over K).

Ax 3: This follows by F_3.

Ax 4: This follows by F_4.

Ax 5: This follows by F_5, as K is transitive. This is because, as K is transitive, the set y defined by

$$(\exists y)(\forall z)(z \in y \equiv (\exists w)(z \in w \wedge w \in x))$$

is precisely the union of x.

Ax 6: This follows by F_6. For, if $x \in K$, then $\mathcal{P}(x) \cap K \in K$. But $\mathcal{P}(x) \cap K$ is precisely the y defined by the formula

$$(\exists y)(\forall z)(z \in y \equiv z \subseteq x)$$

Ax 7: This follows by F_7.

Ax 8: This follows by F_8.

Ax 9: This follows by F_9.

(b) On the other hand, suppose that all the ZF axioms are true over K. We show that K is a first order universe.

F_1: This holds by assumption.

F_2: Consider any $x \in K$ and some first-order definable (over K) subset A of x. By Lemma 1.1, condition F_2' holds, which means that $A \cap x = A$ is in K. Thus K is first-order swelled.

F_3: This follows by **Ax 3** (because K is transitive).

F_4: This follows by **Ax 4** (again because K is transitive).

F_5: This follows by **Ax 5** (again because K is transitive).

F_6: This follows by **Ax 6**.

F_7: This follows by **Ax 7**.

F_8: If F is a function that is first-order definable over K, then it is defined by some formula $\phi(x, y)$. Then, for every $w \in K$, by **Ax 8** there is a $v \in K$ such that

$$(\forall y)(y \in v \equiv (\exists x)(x \in w) \wedge \phi(x, y))$$

which is just saying $F''(w) \in K$.

F_9: This follows by **Ax 9** (because K is transitive).

Exercise 1.2

It is not clear what is being asked here. Are the ZF axioms being discussed the first-order ones, or the ones introduced at the beginning of the book? If they are the former, clearly we cannot be using Theorems from chapters 7 and 8, where the first-order axioms were not yet introduced. For the problem, I am therefore assuming we are speaking of the intial ZF axioms. Well, suppose we add the axiom of well foundedness. By Theorem 3.5 of Chapter 8, it follows that $V = R_\Omega$, which is a Zermelo-Fraenkel Universe by Theorem 4.3 of Chapter 7. Thus these new axioms are consistent by assumption.

This means that the axiom of well foundedness cannot be disproved in ZF for the following reason: For suppose it could be. Then adding the axiom of well foundedness to the ZF axioms would introduce a contradiction, contrary to the result of this exercise.

Section 3

Notes

When we are discussing Lemma 3.2, the formulas $x_i \in x_j$, $x_i \in a$, and $a \in x_i$ are discussed. Using x_i and x_j is possible in the formulas, as they are just two of the denumerably many variables assumed to be available.

The first paragraph of page 177 is confusing. It says that F_7 is the first-order substitution axioms, but it is in reality the axiom of infinity. The book says that $F_1 - F_5$ all follow easily from results of the last chapter, and here is why this is.

F_1: This follows because we know that L is transitive.

F_2: Consider any constructible set x (say it is in some L_α), and any first-order definable (over L) subset y of x. Then $y \in \mathcal{F}(L_\alpha)$ and thus $y \in L_{\alpha+1}$, meaning y is constructible. This means L is first-order swelled.

F_3: Obvious.

F_4: Suppose $x \in L$ and $y \in L$. Suppose α is an ordinal such that both $x \in L_\alpha$ and $y \in L_\alpha$. Then $\{x, y\}$ is first order definable over L_α, meaning $\{x, y\} \in \mathcal{F}(L_\alpha)$ and thus $\{x, y\} \in L_{\alpha+1}$.

F_5: Suppose x is constructible. Then $(\exists y)(z \in y \wedge y \in x)$ defines $\cup x$ over L (because L is transitive).

F_7 also follows as $\omega \in L$. This is because $\omega \in \mathcal{F}(L_\omega)$, as the Num$(x)$ defined in Section 2 of Chapter 12 is absolute, hence absolute over L_ω, and because all natural numbers are in L_ω, Num(x) defines ω over L_ω.

In the paragraph below, when explaining how F_6 follows, here is why $x \subseteq a$ is absolute (over L_α). Recall that $x \subseteq y$ was shown to be absolute in Chapter 12. But what this means is that, for every transitive class K, and any elements a, b of K, $a \subseteq b$ is absolute over K. But this obviously implies that $x \subseteq a$ is also absolute (over L_α) (for, this would happen if, for any element b of L_α, $b \subseteq a$ is absolute over L_α).

Section 4

Notes

It is remarked that the sentence $(\forall x)(\exists y)(\text{Ordinal}(y) \wedge \sim(y \in x))$ is a theroem of ZF. No justification is given for this, though the proof would supposedly be pretty much identical to that given proving that, for any set of ordinals, there is some ordinal outside of the set (because, ultimately, the ZF axioms considered here in formula form are basically exactly the same as the less formal ones given at the beginning of the book, which we used to prove that result).